



# SAFETY DATA SHEET

According to WHS Regulations

Revision date 16-Feb-2022

Revision Number 1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

**Product Name** Salmonella paratyphi A Antigen H (AH) (50 mL)

**Catalogue Number(s)** 63322

### Other means of identification

**Pure substance/mixture** Mixture

### Recommended use of the chemical and restrictions on use

**Recommended use** Restricted to professional users  
In-vitro laboratory reagent or component\*\*\*

**Uses advised against** No information available

### Details of manufacturer or importer

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com\*\*\*

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pty Ltd  
Level 5  
446 Victoria Road,  
Gladesville NSW 2111  
Australia\*\*\*

### For further information, please contact

**Technical Service** +61 2 9914 2800 or 1800 224 354  
sales.australia@bio-rad.com\*\*\*

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Australia: 61-290372994\*\*\*

**Emergency telephone number** No information available

## SECTION 2: Hazards identification

### GHS Classification

Not classified\*\*\*

### Label elements

#### **Hazard statements**

Not classified\*\*\*

#### **Other hazards which do not result in classification**

**General Hazards** Contains human source material and / or potentially infectious components\*\*\*

**SECTION 3: Composition/information on ingredients****Substance**

Not applicable\*\*\*

**Mixture \*\*\***

| Chemical name                 | CAS No          | Weight-%     |
|-------------------------------|-----------------|--------------|
| Formaldehyde***               | 50-00-0         | 0.01 - 0.099 |
| Non-hazardous ingredients *** | Proprietary *** | Balance ***  |

**SECTION 4: First aid measures****Description of first aid measures**

|                                   |                                                                                                            |
|-----------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>             | No hazards which require special first aid measures.                                                       |
| <b>Emergency telephone number</b> | Poisons Information Centre, Australia: 13 11 26<br>Poisons Information Centre, New Zealand: 0800 764 766   |
| <b>Inhalation</b>                 | Remove to fresh air.                                                                                       |
| <b>Eye contact</b>                | Call a doctor. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.*** |
| <b>Skin contact</b>               | Wash with soap and water.***                                                                               |
| <b>Ingestion</b>                  | Call a doctor.***                                                                                          |

**Most important symptoms and effects, both acute and delayed**

|                 |                           |
|-----------------|---------------------------|
| <b>Symptoms</b> | No information available. |
|-----------------|---------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                        |                        |
|------------------------|------------------------|
| <b>Note to doctors</b> | Treat symptomatically. |
|------------------------|------------------------|

**SECTION 5: Firefighting measures****Suitable Extinguishing Media**

|                                     |                                                                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|

|                                       |                           |
|---------------------------------------|---------------------------|
| <b>Unsuitable extinguishing media</b> | No information available. |
|---------------------------------------|---------------------------|

**Specific hazards arising from the chemical**

|                                                   |             |
|---------------------------------------------------|-------------|
| <b>Specific hazards arising from the chemical</b> | None known. |
|---------------------------------------------------|-------------|

**Special protective actions for fire-fighters**

|                                         |                                                                                           |
|-----------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Special protective equipment for</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout |
|-----------------------------------------|-------------------------------------------------------------------------------------------|

fire-fighters gear. Use personal protection equipment.

## SECTION 6: Accidental release measures

### Personal precautions, protective equipment and emergency procedures

**Personal precautions** See section 8 for more information.

**For emergency responders** Use personal protection recommended in Section 8.

### Environmental precautions

**Environmental precautions** See Section 12 for additional Ecological Information.

### Methods and material for containment and cleaning up

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Clean contaminated surface thoroughly.\*\*\*

### Precautions to prevent secondary hazards

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## SECTION 7: Handling and storage

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

### Conditions for safe storage, including any incompatibilities

**Storage Conditions** Store according to product and label instructions.\*\*\*

**Incompatible materials** None known based on information supplied.

## SECTION 8: Exposure controls/personal protection

### Control parameters

**Exposure Limits** \*\*\*

| Chemical name              | Australia                                                                  | ACGIH TLV                     |
|----------------------------|----------------------------------------------------------------------------|-------------------------------|
| Formaldehyde***<br>50-00-0 | 1 ppm<br>1.2 mg/m <sup>3</sup><br>2 ppm STEL<br>2.5 mg/m <sup>3</sup> STEL | STEL: 0.3 ppm<br>TWA: 0.1 ppm |

### Biological occupational exposure limits

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Appropriate engineering controls

**Engineering controls**                      Showers  
                                                      Eyewash stations  
                                                      Ventilation systems.

#### Individual protection measures, such as personal protective equipment

**Eye/face protection**                      Wear safety glasses with side shields (or goggles).\*\*\*

**Skin and body protection**                      Wear suitable protective clothing.\*\*\*

**Hand protection**                              Wear suitable gloves.\*\*\*

**Respiratory protection**                      No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**Environmental exposure controls**                      No information available.

### **SECTION 9: Physical and chemical properties**

#### Information on basic physical and chemical properties

**Physical state**                                  Liquid\*\*\*

**Appearance**                                  Liquid\*\*\*

**Colour**                                          yellow\*\*\*

**Odour**                                          Low.\*\*\*

**Odour threshold**                                  No information available

| <u>Property</u>                               | <u>Values</u>        | <u>Remarks • Method</u> |
|-----------------------------------------------|----------------------|-------------------------|
| <b>pH</b>                                     |                      | None known              |
| <b>Melting point / freezing point</b>         | No data available    | None known              |
| <b>Boiling point / boiling range</b>          | No data available    | None known              |
| <b>Flash point</b>                            | No data available    | None known              |
| <b>Evaporation rate</b>                       | No data available    | None known              |
| <b>Flammability (solid, gas)</b>              | No data available    | None known              |
| <b>Flammability Limit in Air</b>              |                      | None known              |
| <b>Upper flammability or explosive limits</b> | No data available    |                         |
| <b>Lower flammability or explosive limits</b> | No data available    |                         |
| <b>Vapour pressure</b>                        | No data available    | None known              |
| <b>Vapour density</b>                         | No data available    | None known              |
| <b>Relative density</b>                       | No data available    | None known              |
| <b>Water solubility</b>                       | Miscible in water*** | ***                     |
| <b>Solubility(ies)</b>                        | No data available    | None known              |
| <b>Partition coefficient</b>                  | No data available    | None known              |
| <b>Autoignition temperature</b>               | No data available    | None known              |
| <b>Decomposition temperature</b>              |                      | None known              |
| <b>Kinematic viscosity</b>                    | No data available    | None known              |
| <b>Dynamic viscosity</b>                      | No data available    | None known              |
| <b>Explosive properties</b>                   | Not applicable       |                         |
| <b>Oxidising properties</b>                   | Not applicable       |                         |
| <u><b>Other information</b></u>               |                      |                         |
| <b>Molecular weight</b>                       | Not applicable       |                         |
| <b>VOC Content (%)</b>                        | Not applicable       |                         |

### **SECTION 10: Stability and reactivity**

#### Reactivity

**Reactivity** No information available.

#### Chemical stability

**Stability** Stable under normal conditions.

#### Explosion data

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

#### Possibility of hazardous reactions

**Possibility of hazardous reactions** None under normal processing.

#### Conditions to avoid

**Conditions to avoid** None known based on information supplied.

#### Incompatible materials

**Incompatible materials** None known based on information supplied.

#### Hazardous decomposition products

**Hazardous decomposition products** None known based on information supplied.

### **SECTION 11: Toxicological information**

#### Acute toxicity

#### Information on likely routes of exposure

##### **Product Information**

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available

**Symptoms** No information available.

#### Numerical measures of toxicity - Product Information

\*\*\*

| Chemical name   | Oral LD50           | Dermal LD50            | Inhalation LC50          |
|-----------------|---------------------|------------------------|--------------------------|
| Formaldehyde*** | = 100 mg/kg ( Rat ) | = 270 mg/kg ( Rabbit ) | = 0.578 mg/L ( Rat ) 4 h |

See section 16 for terms and abbreviations

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Skin corrosion/irritation** Based on available data, the classification criteria are not met.

**Serious eye damage/eye irritation** Based on available data, the classification criteria are not met.

**Respiratory or skin sensitisation** Based on available data, the classification criteria are not met.

**Germ cell mutagenicity** Based on available data, the classification criteria are not met.

**Carcinogenicity** \*\*\*

The table below indicates whether each agency has listed any ingredient as a carcinogen.\*\*\*

| Chemical name             | Australia |
|---------------------------|-----------|
| Formaldehyde*** - 50-00-0 | Carc. 1B  |

**Reproductive toxicity** Based on available data, the classification criteria are not met.

**STOT - single exposure** Based on available data, the classification criteria are not met.

**STOT - repeated exposure** Based on available data, the classification criteria are not met.

**Aspiration hazard** Based on available data, the classification criteria are not met.

## SECTION 12: Ecological information

### Ecotoxicity

**Ecotoxicity** \*\*\*

**Unknown aquatic toxicity** 0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment.\*\*\*

| Chemical name   | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                                                 | Toxicity to microorganisms | Crustacea                                                                     |
|-----------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------|
| Formaldehyde*** | -                    | LC50: 0.032 - 0.226mL/L (96h, Oncorhynchus mykiss)<br>LC50: 100 - 136mg/L (96h, Oncorhynchus mykiss)<br>LC50: 22.6 - 25.7mg/L (96h, Pimephales promelas)<br>LC50: 23.2 - 29.7mg/L (96h, Pimephales promelas)<br>LC50: =1510µg/L (96h, Lepomis macrochirus)<br>LC50: =41mg/L (96h, Brachydanio rerio) | -                          | EC50: 11.3 - 18mg/L (48h, Daphnia magna)<br>LC50: =2mg/L (48h, Daphnia magna) |

### Persistence and degradability

**Persistence and degradability** No information available.

### Bioaccumulative potential

**Bioaccumulation** There is no data for this product.\*\*\*

**Component Information** \*\*\*

| Chemical name   | Partition coefficient |
|-----------------|-----------------------|
| Formaldehyde*** | 0.35***               |

### Mobility

**Mobility in soil** No information available.

**Mobility** No information available.

#### Other adverse effects

**Other adverse effects** No information available.

### SECTION 13: Disposal considerations

#### Waste treatment methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.\*\*\*

**Contaminated packaging** Do not reuse empty containers.

### SECTION 14: Transport information

**ADG** Not regulated

**IATA** Not regulated\*\*\*

**IMDG** Not regulated\*\*\*

#### **Transport in bulk according to Annex II of MARPOL and the IBC Code**

No information available

### SECTION 15: Regulatory information

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

##### National regulations

##### Australia

See section 8 for national exposure control parameters

##### **Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)**

Classified as a scheduled poison according to the Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP)\*\*\*

**Poison Schedule Number** 6\*\*\*

##### **Major hazard (accident/incident planning) regulation**

Verify that licence requirements are met

Named hazardous chemicals\*\*\*

| Chemical name             | Threshold quantity (T) |
|---------------------------|------------------------|
| Formaldehyde*** - 50-00-0 | 50 tonne TQ >90%       |

\*\*\*

##### **National pollutant inventory**

Subject to reporting requirement\*\*\*

| Chemical name             | National pollutant inventory     |
|---------------------------|----------------------------------|
| Formaldehyde*** - 50-00-0 | 10 tonne/yr Threshold category 1 |

##### International Inventories

Contact supplier for inventory compliance status

### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

## **SECTION 16: Other information**

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 16-Feb-2022

**Revision Note** Significant changes throughout SDS. Review all sections.

### **Key or legend to abbreviations and acronyms used in the safety data sheet**

#### Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

### **Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

### **Disclaimer**

**The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text**

**End of Safety Data Sheet**